

**Johnson C. Smith University**  
**Finite Math | MTH 132 | Section K | Fall 2012**  
**Tuesday 1:30-2:45 | MAIN SHB 201**

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*“Every act of conscious learning requires the willingness to suffer an injury to one’s self-esteem. That is why young children, before they are aware of their own self-importance, learn so easily; and why older persons, especially if vain or important, cannot learn at all.”*

–Thomas Szasz, author, professor of psychiatry (1920- 2012)

## Instructor Information

**Name:** Dr. James Ashe

**Office location:** 307 Davis Hall (back)

**Email:** [jashe@jcsu.edu](mailto:jashe@jcsu.edu)

**Office phone:** (704) 378-1037

**Office hours:** TR 8:30-9:45 and 10:45-12; TR 4:15-7:45 (by appointment)

## Course Description

This is a 3 credit hour course. Elements of finite mathematical systems for liberal arts and education students. Topics include real numbers, linear equations and straight lines, systems of linear equations and inequalities, matrix algebra, sets and counting, concepts of probability and statistics, and mathematics of finance. The course relies heavily on computers and graphing calculators to develop intuition, make estimates, verify results, and check reasonableness of answers. **Prerequisite: MTH 131.**

This is a *hybrid course*. Hybrid courses meet approximately 49% online and 51% face-to-face. Online materials will be disseminated through *Jenzabar* <https://my.jcsu.edu/ics>.

Instructional methods will primarily consist of lectures (via online videos), in-class discussions, and projects. However other methods may be adopted according to the particular needs of the class.

## Course Materials

**Textbook/MyMathLab:** A MyMathLab access code is required. New textbooks and Ebooks usually come packaged with a MyMathLab code. Course ID: ashe08469.

Register and purchase a code here: <http://www.pearsonmylabandmastering.com/northamerica/mymathlab/students/get-registered/index.html>

An access code can also be purchased here:

<http://www.amazon.com/MyMathLab-Student-Hall-Pearson-Education/dp/032119991X>

**Calculator:** TI-83, TI-83+, and TI-84+ are allowed and supported. Other calculators can be used with Instructor permission. Students should bring their calculators to class.

## Grades

Grades will be taken from homework/quizzes, exams, a project, a common Midterm, and common Final as follows:

**Evaluation:** HW/Quizzes = 20%, Exams 1, 2, 3, 4, and Midterm = 12% each; Final Exam = 20%.

**Grading scale:**  $A \geq 90 > B \geq 80 > C \geq 70 > D \geq 60 > F$

**Quizzes:** HW/Quizzes will consist of any graded assignment which is not an exam. This includes a *project based learning module*, traditional quizzes, MyMathLab assignments, take home assignments, and in-class projects. The two lowest quiz grade will be dropped.

**Final Exam and Midterm:** Both the final and midterm exam will be common comprehensive exams given in-class; see the exam schedule [jcsu.edu/happenings/exam-schedule](http://jcsu.edu/happenings/exam-schedule). *If beneficial, the final exam score will replace the lowest exam score.*

## Class Policies

JCSU policies regarding attendance, academic integrity, grades, and change of enrollment will be enforced. It is the student's responsibility to be aware of these policies and of related deadlines as well as any announcements or syllabus adjustments (written or oral) made in class. Please see the student handbook for details, [jcsu.edu/docs/handbook.pdf](http://jcsu.edu/docs/handbook.pdf).

**Attendance:** Johnson C. Smith University has no official attendance policy. However, because attendance in classes is a vital part of the educational process, students are encouraged to attend classes regularly and promptly.

Students who may miss classes while representing the University in an official capacity are exempt from regulations governing absences. However, absence from class for official University business does not relieve the student from responsibility for any class assignments that may be missed during the period of absence.

**Missed quizzes:** There will be no make-ups for in-class quizzes or assignments unless the class was missed for a University sponsored event, and advance notice of the absence was given by the student. In this case the assignment will be excused or a make-up will be given depending on circumstances. Late take-home assignments will have 25% deducted for each day it is late.

**Missed exams:** If a student misses an exam for a verifiable excuse, they should send me an e-mail *as soon as possible* so that a make-up can be arranged. Make-up exams may be significantly more difficult. If you will be gone for any reason, it will be *your* responsibility to inform me at least a week in advance to schedule to take the exam early.

**Student support services:** If you need course adaptations or accommodations for a documented disability please contact the Office of Disability Services.

[jcsu.edu/directory/federal-requirements/general-institutional-information/jcsu-disability-services](http://jcsu.edu/directory/federal-requirements/general-institutional-information/jcsu-disability-services),  
phone: (704) 398-1116.

The Office of Counseling Services provides confidential intakes and assessments, personal counseling and referrals to appropriate resources to all enrolled students on a walk-in or appointment basis.

[jcsu.edu/admissions/future\\_students/meet\\_your\\_counselor](http://jcsu.edu/admissions/future_students/meet_your_counselor), phone (704) 378-1044.

## Course Content

### Chapter 1: Linear Functions

- 1.1 Slopes and Equations of Lines
- 1.2 Linear Functions & Applications
- 1.3 Least Squares Line

### Chapter 2: Systems of Linear Equations

- 2.1 The Echelon Method
- 2.3 The Gauss-Jordan Method
- 2.4 Adding & Subtracting Matrices
- 2.5 Matrix Inverses

### Chapter 5: Math of Finances

- 5.1 Simple and Compound Interest
- 5.2 Future Value of Annuity
- 5.3 Present Value & Amortization of Annuity

### Chapter 6: Logic

- 6.1 Statements
- 6.2 Truth Tables & Equivalent Statements
- 6.4 More on the Conditional

### Chapter 7: Sets and Probability

- 7.1 Sets
- 7.2 Introduction to Probability
- 7.3 Basics of Probability

## Important Dates

|                           |                        |
|---------------------------|------------------------|
| January 17 <sup>th</sup>  | Last day to Add/Drop   |
| January 28 <sup>th</sup>  | Exam 1                 |
| February 18 <sup>th</sup> | Exam 2                 |
| March 4 <sup>th</sup>     | Midterm Exam           |
| April 1 <sup>st</sup>     | Exam 3                 |
| April 15 <sup>th</sup>    | Exam 4                 |
| May 2 <sup>nd</sup>       | Common Final (10-12pm) |

The above exam dates are subject to change. JCSU's academic calendar can be found here: [jcsu.edu/happenings/academic-calendar](http://jcsu.edu/happenings/academic-calendar), and the exam schedule can be found here: <http://www.jcsu.edu/happenings/exam-schedule/spring-exam>

**Disclaimer:** While I do not foresee any significant changes to the items in this syllabus during the semester, I reserve the right to make appropriate changes. Students will be notified of any such changes in writing.